

Roll No.....

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: B.Tech.

Semester: II

Name of the Course: Engineering Mathematics-II
TMA-201

Course Code:

Time: 90 Minutes

Maximum

Marks: 50

Note:

- (i) Answer **all the questions** by choosing *any one of the sub questions*.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO 1
a) Solve:	$\{2xy \cos x^2 - 2xy + 1\}dx + \{\sin x^2 - x^2 + 3\}dy = 0.$	
OR		
b) Solve the following differential equation.	$x \frac{dy}{dx} + y \log y = xye^x$	
Q2	(10 marks)	CO 1
a) Solve the given differential equation:	$\frac{dy}{dx} - \frac{dx}{dy} = \frac{x}{y} - \frac{y}{x}$	
OR		
b) Solve:	$(D^2 - 4D + 3)y = 2xe^{3x} + 3e^{3x} \cos 2x, \text{ where } D = \frac{d}{dx}.$	
Q3	(10 marks)	
a) Solve the given Cauchy Euler differential equation:	$x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = x^2 + \sin(5 \log x)$	

OR		
b)	Solve by method of variation of parameters: $\frac{d^2y}{dx^2} + y = \sec x \tan x$	
Q4		(10 marks)
a)	Prove that: $\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$	CO 2
OR		
b)	Prove that : $J_{\frac{3}{2}}(x) = \sqrt{\frac{2}{\pi x}} \left[\frac{\sin x}{x} - \cos x \right].$	
Q5		(10 marks)
a)	Prove that: $n P_n(x) = x P'_n(x) - P'_{n-1}(x)$	CO 2
OR		
b)	Prove that: $\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1}, \quad \text{if } m = n.$	